

Demo dossier — pulse oximeter

Expected overall result: PASS

SYNTHETIC TEST DOCUMENT — assembled from the MedDevAudit-IN evaluation fixtures for the purpose of exercising the auditor. Not a real submission. Device names, companies, certificate and licence numbers are fictitious.

DEVICE MASTER FILE — INDEX Submitted under Rule 35, Medical Device Rules 2017, Fourth Schedule Part II Device: FingerSense FS-200 Fingertip Pulse Oximeter Applicant: Meridian Medical GmbH, Germany

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SECTION 1 — DEVICE DESCRIPTION The FingerSense FS-200 is a reusable, battery-powered fingertip pulse oximeter that measures functional oxygen saturation (SpO₂) and pulse rate by transmission pulse oximetry at 660 nm (red) and 905 nm (infrared). Maximum optical output power is below 15 mW. The device is intended for spot-check measurement on adult and paediatric patients in professional healthcare facilities and in the home healthcare environment.

SECTION 2 — LICENSING Import licence: MD-15 licence number IMP/MD/2024/004518, issued by the Central Licensing Authority, CDSCO, on 14 March 2024, valid until 13 March 2029. Application submitted in Form MD-14 dated 02 January 2024. Indian Authorised Agent: Sundaram Healthcare Distributors Pvt Ltd, 41 Anna Salai, Chennai 600002, Tamil Nadu, India (wholesale licence 20B/21B No. TN/CH/2019/3312). A notarised power of attorney dated 18 December 2023 appoints Sundaram Healthcare Distributors Pvt Ltd as the authorised agent of Meridian Medical GmbH for the purposes of Rule 35.

Free Sale Certificate: issued by the Bundesinstitut fuer Arzneimittel und Medizinprodukte (BfArM), the national regulatory authority of the country of origin (Germany), certificate 2023/FSC/88214 dated 05 November 2023, confirming that the FS-200 is freely sold in Germany.

SECTION 3 — QUALITY MANAGEMENT SYSTEM ISO 13485:2016 certificate number DE-QMS-2022-77413 issued by TUV SUD Product Service GmbH (accredited certification body). Scope of certification: design, development and manufacture of non-invasive patient monitoring equipment including pulse oximeters. Certified site: Meridian Medical GmbH, Industriestrasse 12, 82256 Fuerstenfeldbruck, Germany. Date of issue 11 April 2022. Valid until 10 April 2026 (date of expiry).

SECTION 4 — RISK MANAGEMENT Risk management has been conducted in accordance with ISO 14971:2019. Documents included: risk management plan RMP-FS200-01; hazard identification and risk analysis worksheet (design FMEA, 64 identified hazards); risk evaluation and risk control measure table with verification of effectiveness; residual risk evaluation; and signed risk management report RMR-FS200-03 concluding that overall residual risk is acceptable in relation to the clinical benefit.

SECTION 5 — SAFETY AND EMC TESTING IEC 60601-1:2005 + A1:2012 + A2:2020 general safety — test report number TR-60601-1-22884 issued by TUV SUD Product Service GmbH, an ILAC-accredited test laboratory. Applied part classified as Type BF. Conclusion: PASS.

IEC 60601-1-2:2014 + A1:2020 electromagnetic compatibility — test report number TR-EMC-22891. Emission tested to CISPR 11 Group 1 Class B. Immunity tested at the higher levels required for the home healthcare environment. The EMC declaration tables are reproduced in Section 8 of the Instructions for Use. Conclusion: PASS.

IEC 60601-1-8:2006 + A1 + A2 alarm system — test report TR-ALM-22903. Alarm conditions: low SpO₂ (high priority), low/high pulse rate (medium priority), low battery and probe-off (technical, low

priority). Audible and visual alarm signal characteristics, default alarm limit settings, alarm limit adjustment range, the alarm-off and audio-paused indications, and the alarm system delay of less than 10 seconds were all verified. Conclusion: PASS.

SECTION 6 — CLINICAL ACCURACY (SpO₂) A controlled desaturation study was performed in accordance with ISO 80601-2-61:2017 clause 201.12.1.101 on 12 healthy adult volunteers, yielding 216 paired data points against a CO-oximeter reference over the saturation range 70 to 100 percent. The subject cohort included 4 subjects with dark skin pigmentation (Fitzpatrick V-VI), 4 with intermediate and 4 with light pigmentation. Result: SpO₂ accuracy root mean square (ARMS) of 1.9 percent over 70 to 100 percent saturation. Bland-Altman analysis is provided in Annex 6A with a mean bias of -0.3 percent and limits of agreement of -4.1 to +3.5 percent.

SECTION 7 — SOFTWARE The embedded firmware is developed under a software lifecycle process conforming to IEC 62304:2006 + A1:2015. Software safety classification: Class B, justified on the basis that a software failure could contribute to a hazardous situation resulting in non-serious injury through a delayed or erroneous saturation reading. Included: software development plan SDP-FS200; software requirements specification SRS-FS200 rev 4; software architecture document; software verification and validation report SVVR-FS200; anomaly list with residual anomaly justification; and the SOUP inventory (FreeRTOS 10.4.3, CMSIS-DSP 1.9.0) with its associated risk assessment.

SECTION 8 — LABELLING ARTWORK (carton and device) Manufactured by: Meridian Medical GmbH, Industriestrasse 12, 82256 Fuerstenfeldbruck, Germany Imported and marketed by: Sundaram Healthcare Distributors Pvt Ltd, 41 Anna Salai, Chennai 600002, India Import licence number: IMP/MD/2024/004518 Model: FS-200 Catalogue number (REF): FS200-IN Batch/Lot number: LOT _____ Serial number: SN _____ Date of manufacture: 2024-____ Use by: not applicable (reusable) Declaration: "For Sale in India" / Hindi (Devanagari) equivalent printed immediately below in bilingual format, together with the bilingual Hindi and English statement of intended use. Symbols used on the artwork conform to ISO 15223-1:2021. Symbol glossary (legend) reproduced in the Instructions for Use section 1: manufacturer, date of manufacture, batch code, serial number, catalogue number, consult instructions for use, Type BF applied part. Declared performance on the label: SpO₂ accuracy ARMS 1.9 percent (70-100 percent); pulse rate range 30-250 bpm, pulse rate accuracy plus or minus 2 bpm; wavelengths 660 nm and 905 nm, maximum optical output power below 15 mW. Compatible sensor/probe list: FS-ADT-01 (adult), FS-PED-01 (paediatric). Use of any probe not on this list invalidates the declared accuracy.

SECTION 9 — INSTRUCTIONS FOR USE (supplied in English) Contents: intended use and intended user; indications for use; step-by-step operating instructions; contraindications; warnings and precautions (including the warning that the declared accuracy statistic is not the measurement uncertainty of an individual reading, and that nail polish, poor perfusion, motion artefact and carboxyhaemoglobin affect readings); residual risks communicated to the user; cleaning and maintenance instructions; storage and transport conditions (-20 to +60 degrees C, 10 to 93 percent RH); expected service life of 5 years; and the contact details for reporting an adverse event or complaint in India (Materiovigilance cell, Sundaram Healthcare Distributors Pvt Ltd, vigilance@sundaramhealth.in, +91 44 4000 1200).

SECTION 10 — POST-MARKET SURVEILLANCE Post-market surveillance plan PMS-FS200 rev 2, including the periodic safety update schedule, complaint handling procedure, and the vigilance reporting route to CDSCO under the Materiovigilance Programme of India.